

NEPHROLOGY TEACHING SERIES

# Hypertension

*Outpatient approach and resistant HTN*

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Home BP data outweighs office readings. Never escalate from one office measurement.

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*MS3/MS4 rotation teaching*

# Why it matters

*Common, silent, modifiable — and still the leading modifiable CV risk factor.*

## WHAT THE CLINIC VISIT COVERS

Confirm with home / ambulatory BP

Target: < 130/80; < 120 in CKD / high CV risk

Lifestyle, then staged 3-drug ladder

Adherence + side effects first

Secondary causes in resistant HTN

CV risk: lipid, DM, smoking

End-organ: UACR, ECG (LVH), fundus

## TEACHING PEARL

*Office BP is a hypothesis, not a diagnosis.  
Check adherence, cuff size, home log before  
adding drugs. Most 'resistant' HTN is  
pseudo-resistant.*

# Diagnosis and targets

*Measurement technique drives both.*

| Category            | SBP / DBP (office) | Home average   | Action                                          |
|---------------------|--------------------|----------------|-------------------------------------------------|
| Normal              | < 120 / < 80       | < 120 / < 80   | Lifestyle; recheck annually                     |
| Elevated            | 120–129 / < 80     | 120–129 / < 80 | Lifestyle; recheck 3–6 mo                       |
| Stage 1             | 130–139 / 80–89    | ≥ 130 / ≥ 80   | Lifestyle ± drug if CV risk ≥ 10%               |
| Stage 2             | ≥ 140 / ≥ 90       | ≥ 135 / ≥ 85   | Drug + lifestyle; 2-drug combo if ≥ 160/100     |
| Hypertensive crisis | ≥ 180 / ≥ 120      | —              | Urgent evaluation — end-organ damage? IV if yes |

**Targets by population:** General < 130/80 (ACC/AHA) • CKD standardized office SBP < 120 when tolerated (KDIGO 2021) • Elderly consider tolerability • Orthostasis → aim at standing SBP too.

# Measurement — the most common error

*Technique changes SBP by 10–20 mmHg. Do it right before escalating.*

## OFFICE BP

5 minutes quiet seating; feet flat on floor; no caffeine / nicotine / exercise 30 min prior

Correct cuff size: bladder width 40%, length 80% of arm circumference

Back supported, arm at heart level, uncovered

No talking during measurement

Average 2 readings at least 1 min apart

**Automated office BP (AOBP) = unattended, automated; most aligned with SPRINT**

## HOME / AMBULATORY BP

Gold standard for diagnosis & titration

7 days of morning + evening readings, duplicates, discard day 1

Validated upper-arm cuff (not wrist)

Home BP averages are 5 mmHg LOWER than office — use HBP thresholds

**White-coat HTN: elevated office, normal home → monitor only**

**Masked HTN: normal office, elevated home → treat**

# The drug ladder

*Three classes, titrated to target. Add spironolactone if resistant.*

| Step | Agent                                                  | Preferred when                                | Avoid when                                               |
|------|--------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------|
| 1    | ACEi or ARB (not both)                                 | CKD (UACR > 30), DM, HFrEF, post-MI           | Pregnancy; bilateral renal artery stenosis (RAS); hyperK |
| 2    | Thiazide (chlorthalidone 12.5–25 mg preferred)         | Older patient, Black patients, low-renin HTN  | eGFR < 30 (use loop instead), gout                       |
| 3    | DHP calcium channel blocker (CCB) (amlodipine 5–10 mg) | Elderly isolated systolic HTN, Black patients | HFrEF (non-DHP), bradyarrhythmia (non-DHP)               |
| 4    | Spironolactone 25–50 mg (or finerenone in DKD)         | Resistant HTN on max 3-drug combo (PATHWAY-2) | K > 5.0, severe CKD; watch gynecomastia                  |
| +    | Beta-blocker, clonidine, hydralazine                   | Add after 4th drug if still not controlled    | Assess for secondary HTN at this point                   |

# Secondary hypertension — who to work up

*Not a shotgun panel. Pick the test based on the clue.*

| Clue                                                  | Suspect                                       | Test                                                                                        |
|-------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------|
| Hypokalemia (spontaneous or diuretic-exaggerated)     | Primary aldosteronism                         | Aldosterone/renin ratio (morning; OK on most BP meds except an MRA; correct low K first)    |
| HTN in young adult or resistant, with renal asymmetry | Fibromuscular dysplasia / atherosclerotic RAS | Duplex renal US → CTA/MRA                                                                   |
| Flash pulmonary edema, acute Cr bump on ACEi          | Bilateral RAS                                 | Renal duplex / MRA                                                                          |
| Paroxysmal HTN + headache, sweating, palpitations     | Pheochromocytoma                              | Plasma metanephrines (free, fractionated)                                                   |
| Moon facies, central obesity, striae, DM, hypoK       | Cushing syndrome                              | 24-h urinary free cortisol; low-dose dexamethasone suppression test (LDDST); 11 pm salivary |
| Snoring, daytime sleepiness, BMI > 30                 | Obstructive sleep apnea                       | Home sleep study → PSG                                                                      |
| HTN in pregnancy                                      | Preeclampsia, chronic HTN                     | Urgent OB; ACEi/ARB contraindicated                                                         |
| Young HTN + weak femoral pulses / rib notching        | Coarctation                                   | CXR, echo, CTA/MRA                                                                          |

# Resistant hypertension workup

*True resistant HTN / pseudo-resistant. Rule out the latter first.*

## PSEUDO-RESISTANT (check first)

Non-adherence (the #1 cause)  
White-coat effect — confirm with home/ambulatory BP  
Sub-optimal drug dosing (not titrated to max)  
Wrong cuff size; improper technique  
Volume overload (high Na diet, inadequate diuretic)  
Interfering substances: NSAIDs, OCPs, pseudoephedrine, alcohol, stimulants, herbal

**If ruled out → true resistant; now pursue secondary workup**

## TRUE RESISTANT HTN — STEPS

Definition: BP > 130/80 on 3 maximized meds (incl. diuretic); OR controlled on 4 meds

**Step 1: add spironolactone 25 mg (PATHWAY-2: best in resistant HTN)**

Check K<sup>+</sup> at 2 weeks; titrate to 50 mg if tolerated

**Step 2: work up secondary HTN comprehensively**

Step 3: add  $\beta$ -blocker, central  $\alpha$  agonist, hydralazine

Renal denervation (investigational, limited) — SPYRAL/RADIANCE trials

Consider direct-acting vasodilators, nitroprusside only in crisis

# Landmark HTN trials

*The RCTs that set modern targets and drug choices.*

| Trial     | Year | Takeaway                                                                            |
|-----------|------|-------------------------------------------------------------------------------------|
| ALLHAT    | 2002 | Chlorthalidone as good as ACEi or CCB for HTN — cheaper first-line option.          |
| HOPE      | 2000 | Ramipril ↓ CV events in high-risk non-HTN patients — BP-independent benefit.        |
| ACCORD-BP | 2010 | Intensive BP (< 120) in T2DM — no CV benefit, more AE. Called SPRINT into question. |
| SPRINT    | 2015 | SBP < 120 (by AOBP) ↓ CV 25% and mortality 27% vs < 140 in non-DM high-risk.        |
| PATHWAY-2 | 2015 | Spironolactone > bisoprolol, doxazosin for resistant HTN. NNT = 4.                  |
| HOT       | 1998 | DBP target: < 80 = < 85 = < 90 in most, but DM benefited from < 80.                 |
| SYST-EUR  | 1997 | CCB reduced stroke in elderly isolated systolic HTN.                                |
| STEP      | 2021 | Intensive (< 130) vs standard (< 150) in elderly Chinese — 26% ↓ CV events.         |

# Common clinic mistakes

*The errors that lead to under-control or over-treatment.*

## AVOID

- Escalating from one office reading
- Prescribing without home BP log
- Ignoring adherence
- ACEi + ARB (ONTARGET)
- Thiazide at eGFR < 30
- Blaming ACEi for 10% Cr rise
- Missing OSA, NSAIDs, decongestants

## DO INSTEAD

- 7-day home log first
- Pill count + open adherence Qs
- Check cuff size + technique
- One RAAS blocker, max dose
- Loop if eGFR < 30
- Accept Cr ↑ up to 30% on ACEi
- Always ask about OSA + OTCs

# Clinical case

*Think it through — answer is on the next slide.*

## VIGNETTE

A 52-year-old woman with T2DM, BMI 33, snoring. On lisinopril 40, HCTZ 25, amlodipine 10, metoprolol 50 BID. Office BP today 152/92 (Korotkoff, repeat 150/90). Home BP log: 148/88 to 155/94 am, 140/85 pm. Adherent by pill-count. K<sup>+</sup> 3.4, spontaneously.

**QUESTION:** Is this resistant HTN? What's the most likely secondary cause and next step?

# Clinical case — answer

*Key teaching points from the case.*

## ANSWER

**Yes (uncontrolled on 3 appropriate agents incl. diuretic). Primary aldosteronism — spontaneous hypokalemia + resistant HTN. Check aldosterone/renin ratio (morning; she can stay on her current meds; correct K first).**

## TEACHING

True resistant HTN requires 3 appropriately-dosed agents including a diuretic. Before adding a 4th drug, always work up secondary causes. Primary aldosteronism is the #1 secondary cause in resistant HTN, especially if spontaneous or diuretic-exaggerated hypokalemia. Testing (Endocrine Society 2025): AM aldosterone/renin ratio on current meds except an MRA; correct K first. A positive screen on ACEi/ARB/diuretic strongly suggests PA; if negative but suspicion stays high, retest after 4 wk off MRA/diuretic/amiloride and ~2 wk off ACEi/ARB/DHP CCB (bridge with verapamil or doxazosin). Other secondary causes: OSA (she snores!), RAS (younger women → FMD), pheo, Cushing, CKD, CoA. Next step after workup: add spironolactone 25 mg (PATHWAY-2 NNT 4 for resistant HTN). Also: weight loss, sleep study.

## REFERENCES

# Hypertension

### Guidelines

ACC/AHA 2017 Hypertension Guideline (+ 2025 update)

KDIGO 2021 Blood Pressure in CKD

Endocrine Society 2016/2025 Primary Aldosteronism

AHA 2018 Resistant Hypertension Scientific Statement

AHA 2019 BP Measurement in Humans

### Landmark Trials

ALLHAT — JAMA 2002

HOPE — NEJM 2000

ACCORD-BP — NEJM 2010

SPRINT — NEJM 2015

PATHWAY-2 — Lancet 2015

HOT — Lancet 1998

SYST-EUR — Lancet 1997

STEP — NEJM 2021

### UpToDate (2026 access):

*Overview of hypertension in adults • Choice of drug therapy in primary (essential) hypertension • Evaluation of resistant hypertension • Primary aldosteronism: screening and diagnosis • Ambulatory and home blood pressure monitoring*